

AI4AI4Cell: COLLABORATIVE EVIDENCE-GUIDED RESEARCH FOR BIOLOGICAL MODEL IMPROVEMENT

Anonymous authors

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ABSTRACT

Biological data are often distributed across institutions and data holders, where privacy and data-sharing constraints limit centralized access. A key challenge for self-improving AI in this setting is to use such distributed evidence to improve both model parameters and the computational solutions built around them. Collaborative learning typically optimizes fixed models, whereas automated model design and research generally rely on centralized training and evaluation signals. Bridging the two is nontrivial: distributed evidence must support not only optimization, but also the evaluation and refinement of computational changes that generalize beyond their local settings. To tackle this, we introduce AI4AI4Cell, a collaborative research framework that couples client-local parameter fitting with an evidence-guided proposal–evaluation loop. Local training shards update a shared predictor, while local development shards produce aggregate evidence cards that guide the next executable proposal. Across three biological settings spanning drug–target interaction prediction, observed-response proteomics, and cell perturbation, the complete system leads four of five main-table endpoints. Under matched data, candidate and compute budgets, evidence-guided allocation yields four held-out wins, eight ties and no losses across twelve task–seed comparisons, including a 10.84-point mean Top-1 gain on double-gene perturbations. AI4AI4Cell offers a practical path toward self-improving biological AI under data-sharing constraints.

1. INTRODUCTION

Biological data are often distributed across institutions and data holders, where privacy and data-sharing constraints limit centralized access. Complementary measurements of molecular interactions, protein responses and cellular perturbations could improve predictive models, but cannot always be pooled across laboratories (Bakhtiari et al., 2025; Sheller et al., 2020). A key challenge for self-improving AI in this setting is to use such distributed evidence to improve both model parameters and the computational solutions built around them.

Collaborative learning typically optimizes parameters within a predefined model design through client-local updates (McMahan et al., 2017; Sheller et al., 2020; Wang et al., 2024; Ding et al., 2026), whereas automated model design and research generally rely on centralized training and evaluation signals (Jiang et al., 2025; Lu et al., 2024; Zhou et al., 2026). Bridging the two is nontrivial: distributed evidence must support not only parameter optimization, but also the evaluation and refinement of computational changes that generalize beyond the local settings in which they were proposed. A locally improved fit need not be a better shared predictor, and a computational revision selected by one data holder may fail on measurements held elsewhere. The central question is therefore how local evidence can jointly drive parameter learning and executable proposal refinement.

To tackle this problem, we introduce AI4AI4Cell, a collaborative research framework coupling client-local parameter fitting with an evidence-guided research harness. Its inner training pipeline distributes a candidate predictor to participating laboratories, fits it on local measurements and aggregates the resulting model updates. Its outer loop reads a task contract and the accumulated experiment history, formulates a focused hypothesis, compiles it into an executable model or optimization revision, evaluates the revision across local development shards, and retains the stronger design. Local measurements therefore play two distinct and complementary roles: training shards update predictive parameters, while development shards produce aggregate evidence for selecting the next computa-

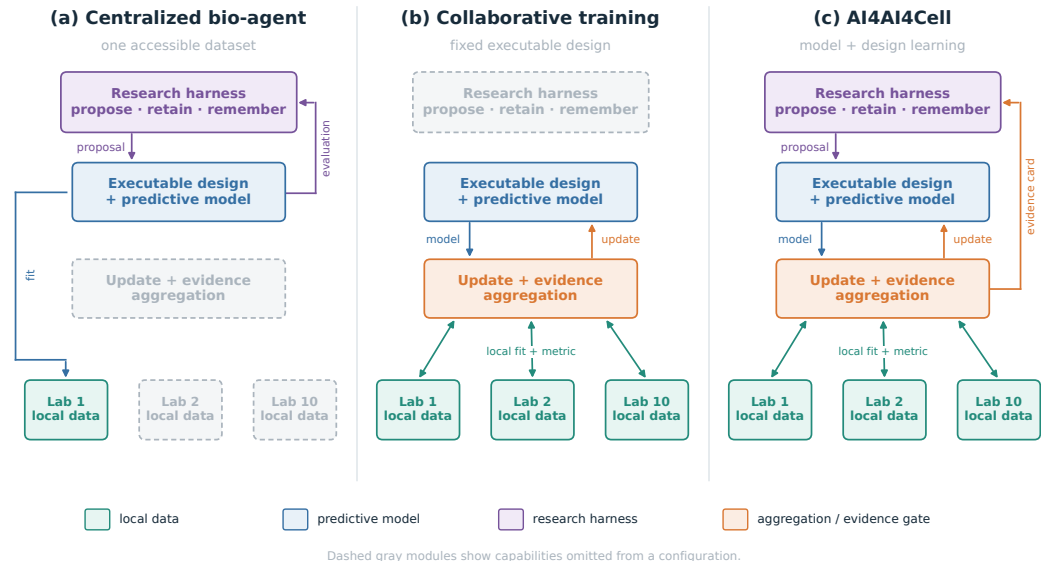


Figure 1: Three research configurations built from the same functional modules. A centralized bio-agent combines an accessible dataset, a predictive model and a research harness. Collaborative training activates multiple local datasets and model-update aggregation while leaving the harness inactive. AI4AI4Cell activates the complete system: local data update the shared model, and aggregate development evidence closes the outer design loop. Ghosted modules mark capabilities omitted from the corresponding configuration.

tional design. Each trial—including rejected and failed trials—adds a compact evidence card to the research history, allowing subsequent proposals to build on what was tried and measured without exposing raw biological responses to the research model. The resulting objects are both a shared predictor and a reproducible design trajectory.

We instantiate the same training pipeline, proposer, candidate library, budget and retention rule in three biological settings: drug–target interaction (DTI) prediction, observed-response proteomic efficacy classification, and five-option cell perturbation identification spanning single-gene, double-gene and drug perturbations. Each study uses ten simulated laboratories. The main comparison contrasts a fixed model or direct optimizer using one laboratory with the complete collaborative research framework using ten, while controlled ablations separate participation, design search and iterative feedback. Our contributions are (i) a collaborative biological research framework that links client-local parameter fitting to distributed evaluation of executable designs; (ii) an evidence-guided harness that structures each proposal as a hypothesis, experiment, expected effect and machine-resolvable design, packages every outcome into persistent history, and uses early evidence to promote computational designs for full validation; and (iii) a common evaluation across three biological task families, including a compute-matched racing experiment that holds data, candidates, total client-epochs and development measurements fixed. Together, these components provide a concrete route for distributed biological evidence to improve both predictors and the computational procedures that produce them.

2. RELATED WORK AND COMPARISON SCOPE

Automated research systems connect computational proposals to measured outcomes. AIDE searches candidate programs through drafting, debugging and improvement conditioned on summarized evaluations (Jiang et al., 2025). The original AI Scientist connects ideas to executed experiments, notes, replanning and writing (Lu et al., 2024). DrugEvolve combines research, implementation, debugging, evaluation, analysis, memory and algorithm lineage across drug-development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). These systems motivate persistent experiment memory and executable revision. AI4AI4Cell extends that research-loop abstraction to measurements distributed across data-owning laboratories: proposal selection is driven by aggre-

gate client evidence, and the same evidence interface is evaluated across DTI, proteomics and cell perturbation.

Collaborative biological training supplies the complementary parameter-learning interface. Medical multi-institutional studies and biological applications such as scFed, FedscGen and Tabula/Chiron use local fitting and parameter aggregation for cell-type classification, batch correction or foundation-model training (Sheller et al., 2020; Wang et al., 2024; Bakhtiari et al., 2025; Ding et al., 2026; Mechtel et al., 2026). AI4AI4Cell uses sample-weighted parameter aggregation (McMahan et al., 2017) as a shared inner training pipeline, then exposes aggregate development evidence to an outer harness that can revise the executable design itself.

Automated design can also interact with local-update training: FedEx tunes client hyperparameters while training shared weights (Khodak et al., 2021), and Helmsman generates and repairs collaborative-learning systems through planning, modular coding and simulation-driven refinement (Li et al., 2025). AI4AI4Cell focuses this combination on biological data holders and contributes a common proposal–measurement–revision interface: the harness records the hypothesis, executable intervention, aggregate outcome and retention decision, then supplies that structured history to the next proposal.

DrugBAN combines molecular graphs, protein sequences and bilinear attention (Bai et al., 2023; DrugBAN contributors, 2026); our DTI study uses the more recent TAPB architecture with public frozen protein features and freshly task-trained parameters (Lin et al., 2025). ProteinTalks supplies the architecture for the observed-response efficacy head (Sun et al., 2026), and scDEBART supplies public-pretrained cell representations (Sung & Kim, 2026). Simple linear models can match or exceed deep models on gene-perturbation response benchmarks (Ahlmann-Eltze et al., 2025), motivating strong fixed recipes and simple references alongside the shared research interface.

3. METHOD: COLLABORATIVE AGENTIC BIOLOGICAL RESEARCH

3.1 RESEARCH OBJECTS AND THE TWO-LEVEL CYCLE

AI4AI4Cell organizes every task around four explicit objects. A *task contract* specifies the primary metric, editable design surface, training budget and data boundary. An *incumbent design* specifies the current model and optimization configuration. A set of *laboratory workers* owns disjoint training and development shards. A persistent *research history* stores one evidence card for every completed, rejected or failed proposal.

Starting from the same task-specific initialization, one research iteration proceeds as follows:

1. The proposer reads the task contract, incumbent design and accumulated history, then emits one structured proposal.
2. A compiler validates the proposal and converts its design identifier into an executable configuration.
3. Each participating laboratory fits that candidate on its local training shard; the coordinator aggregates model updates into a shared predictor.
4. Laboratories evaluate the shared predictor on their local development shards. The coordinator reduces these measurements to an equal-client mean primary score, mean loss and learning-curve diagnostics.
5. The finalizer compares the candidate with the incumbent, records the outcome as an evidence card and carries the selected design into the next iteration.

This cycle makes the two feedback levels concrete. Training measurements change the shared parameters inside a candidate design; development measurements determine which design is retained and what evidence is available to the next research step. The language model remains fixed throughout. Every candidate restarts from the same initialization for its task and seed, so a later proposal does not inherit extra training merely because it was proposed later. Client 0 supplies selection evidence in one-laboratory arms; all ten clients contribute equally weighted selection evidence in collaborative arms.

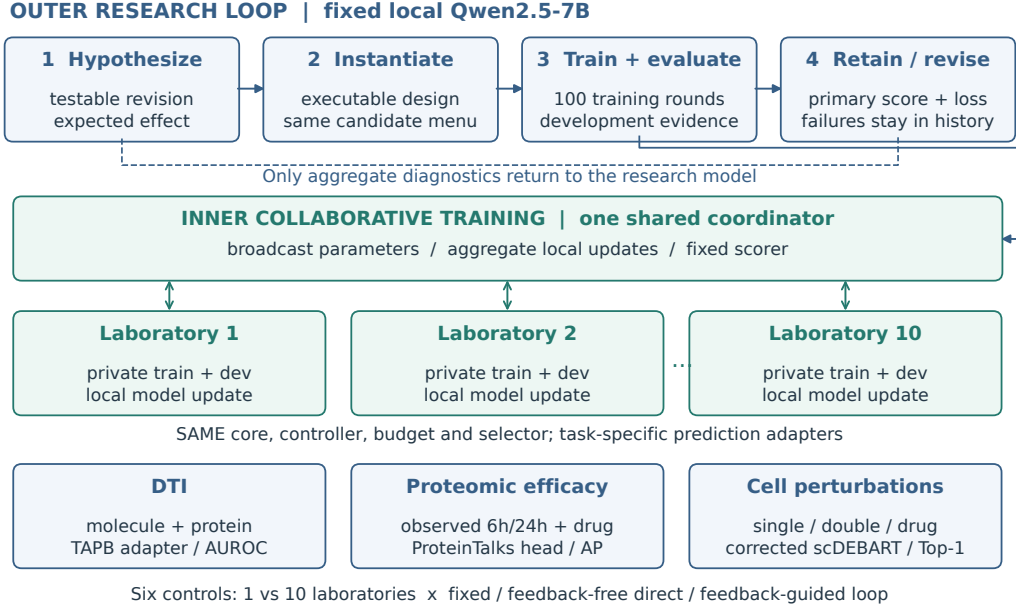


Figure 2: One executable research cycle across three biological task families. The outer harness proposes and compiles a design; the inner training pipeline fits it through data-owning laboratory workers. Aggregated model updates produce the candidate predictor, and aggregate development diagnostics form an evidence card for finalization and the next proposal. Laboratory participation and research feedback are crossed in the six-arm evaluation.

3.2 SHARED INNER TRAINING PIPELINE

At round r , each participating laboratory receives shared parameters w^r and completes one local epoch with AdamW. Local optimizer state resets each round, batch size is 64, and the gradient norm is clipped at one. With local fitted parameters w_i^{r+1} and training-example counts n_i , the shared update is

$$\bar{w}^{r+1} = \sum_{i \in \mathcal{K}} \frac{n_i}{\sum_{j \in \mathcal{K}} n_j} w_i^{r+1}, \quad v^{r+1} = \beta v^r + \bar{w}^{r+1} - w^r, \quad w^{r+1} = w^r + v^{r+1}.$$

Setting $\beta = 0$ gives sample-weighted FedAvg (McMahan et al., 2017). Momentum acts on trainable parameters, floating model buffers are averaged, and integer buffers use an elementwise maximum. This treatment preserves valid local BatchNorm state while keeping the update rule identical across all task adapters. A one-client run uses the same code path with $|\mathcal{K}| = 1$.

Each full-validation candidate follows a 100-round schedule. Every five rounds, local development workers return scalar metrics; the coordinator retains the checkpoint with the highest equal-client mean primary metric, breaking ties by lower mean development loss. Model selection is completed on development evidence, after which the selected checkpoint is frozen and evaluated with the common held-out scorer. The DTI adapter trains TAPB on frozen public protein features; proteomics trains a ProteinTalks-derived efficacy head; cell tasks train the response head on frozen, mask-corrected scDEBART representations. Only these task adapters and their scorers differ across studies.

3.3 STRUCTURED PROPOSALS AND PERSISTENT EVIDENCE

The shared research model is local Qwen2.5-7B-Instruct. Every proposal follows the same four-field JSON contract:

Field	Required content	Operational role
<code>hypothesis</code>	One focused claim about an editable factor	States why the revision should help
<code>experiment</code>	The exact controlled change and comparison	Defines what will be executed
<code>expected_effect</code>	Expected metric direction and diagnostic signature	Makes the proposal testable
<code>design_id</code>	A machine-resolvable candidate identifier	Compiles the proposal to a valid configuration

For example, a proposal may predict that lower learning rate with a residual response adapter improves development ranking without increasing loss, request the corresponding comparison with the incumbent, and select the identifier that implements that combination. The identifier, rather than free-form prose, is the executable source of truth.

In the uniform study, identifiers address a common twelve-design library spanning learning rate, weight decay, server momentum and an optional zero-output-initialized width-32 residual adapter. This bounded library supplies a controlled instance of the broader harness: it preserves hypothesis formation, executable compilation, isolated evaluation, retention and experiment memory while making the search space identical across tasks and research modes. Appendix A lists every design.

After execution, the harness appends an evidence card containing the configuration, aggregate best development metric and loss, selected round, first and last training/development diagnostics, and the accepted, rejected or failed decision. The next loop proposal receives the incumbent plus this complete sequence of evidence cards. Rejected trials remain useful because they identify exhausted directions and observed failure modes. The feedback-free direct arm uses the same Qwen checkpoint, candidate library, six proposal slots and fitting service, but receives only the identifiers it has already tried. Both modes share an identical first proposal, making subsequent divergence attributable to evaluation history rather than a different starting sample.

The finalizer applies the same primary-metric/loss ordering used for checkpoint selection. Completed candidates can replace the incumbent; numerical failures are recorded and advance the proposal budget. Identical configurations within the same seed and participation setting may reuse a verified completed fit while retaining their logical position in both research traces. Proposal generation, fitting and communication costs are recorded separately.

To isolate whether evaluation history has operational value, we additionally instantiate the finalizer as a compute-matched racing loop. Both feedback-free and loop arms receive the same ten-design fractional-factorial slate, covering low/default/high learning rates and interactions among weight decay, server momentum and the residual head. Feedback-free search allocates 80 rounds to every design. The loop first screens every design for 20 rounds, then promotes six designs using two complementary signals within each learning-rate stratum: the best current primary/loss ordering and the largest round-5-to-20 primary-score increase. Promoted designs restart from the common initialization and receive 100-round validation. The two arms therefore use exactly 800 full-client training rounds, 160 development measurements and ten candidate hypotheses; only evidence-dependent allocation differs. This screening–promotion–validation pattern uses the same proposal fields and evidence cards as the general harness, while making the causal role of feedback directly measurable.

3.4 LABORATORY AND COORDINATOR INTERFACES

Laboratory workers read their own training and development shards. During fitting they return model updates and example counts; during evaluation they return scalar metrics and diagnostics. The coordinator distributes the shared predictor, aggregates parameter updates, builds the evidence card and exposes that compact record to the research harness. Consequently, a new laboratory can influence the model through local fitting and influence the design trajectory through local evaluation while its raw profiles or interaction rows remain at the worker interface.

The current experiments instantiate these roles with ten same-host workers and sample-weighted aggregation. The interface is compatible with secure transport, secure aggregation and differential-privacy layers, which can be added independently of the proposal and evidence contracts. The six-arm factorial design evaluates laboratory participation and research feedback as separate experimental factors.

4. EXPERIMENTAL DESIGN

4.1 THREE QUESTIONS: PARAMETERS, PROPOSALS, AND GENERALIZATION

Each case tests whether distributed measurements improve a shared predictor at both the parameter and design levels, and whether retained changes generalize beyond the local setting that produced them. We assign training and development groups to ten disjoint simulated laboratories, balancing by compounds or intervention identities rather than equal cell counts. A single-laboratory method accesses only preselected client 0; a ten-laboratory method uses all ten clients. Within a task, every arm uses the same scorer and candidate rosters.

Benchmark contracts.

Setting	Prediction contract	Task-reference model	Primary metric
DTI	Compound + protein $\rightarrow$ interaction	TAPB	AUROC $\uparrow$
Proteomics	Proteome + drug $\rightarrow$ efficacy	ProteinTalks-derived head	AP $\uparrow$
Cell (3 strata)	Response + 5 options $\rightarrow$ identity	Corrected scDEBART head	Macro Top-1 $\uparrow$

The cell contract is instantiated for VCC single-gene, Norman double-gene and Tahoe drug perturbations; task-specific held-out units and secondary metrics are specified below.

Three complementary contrasts address the two-level question. Ten- versus one-laboratory training, with research mode fixed, measures the value of broader laboratory participation through additional training data and development evidence. Loop versus fixed measures the complete research workflow. To isolate iterative feedback, the compute-matched racing study fixes the ten-design slate, ten-laboratory access, total client-epoch budget and number of development measurements, varying only whether early evidence reallocates later validation compute. After development selection, a common held-out scorer measures transfer of the selected predictor.

The main operational table reports a fixed model and direct optimizer with one laboratory against our complete ten-laboratory harness. A factorial ablation contains all six combinations of one/ten laboratories and fixed/direct/loop training; seeds 42, 43 and 44 are shared across these arms. “Fixed” denotes a fixed architecture and optimization recipe whose predictive parameters still receive the complete training schedule. The feedback-isolation study uses newly executed seeds 53, 54 and 55 for proteomics and cell perturbation. Its direct arm trains all ten candidates for 80 rounds; its loop screens all ten for 20 rounds and fully validates six promoted candidates for 100 rounds, giving both arms 800 full-client rounds and 160 development measurements.

Selection uses participating clients’ development data. After selection, every arm is evaluated on the same ten-client diagnostic panel and frozen held-out test set. Equal-client AUROC/AP gives each laboratory the same influence; the cell clients have equal development-intervention counts, so their mean macro score equals the overall intervention-macro score. Tables aggregate all three prespecified seeds and report sample SD. Checkpoint, configuration and data hashes bind each score to its selected run. Bold marks the best displayed result within each task column, including ties.

4.2 DRUG-TARGET INTERACTION PREDICTION

DTI predicts a binary interaction from a compound and a protein sequence. We use the released BindingDB training/development split of 50,149/5,604 pairs. Canonical-compound groups are assigned to clients with a label-independent size-balancing rule, producing compound-disjoint clients while preserving the source split within each client.

The task reference uses the released TAPB architecture with frozen public ESM2 protein features and freshly initialized drug, fusion and classification modules (Lin et al., 2025). A target dictionary derived from client 0’s training proteins is shared identically by every arm, ensuring a common initialization. The objective combines TAPB’s classification NLL and masked-language loss under the shared batch-64 training pipeline. Candidate designs can add a zero-output residual adapter before TAPB’s native confounder-averaged prediction.

Selection uses mean-client AUROC; AP, MCC, accuracy at 0.5 and binary cross-entropy are secondary. The fixed test suite contains 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein

pairs. Random-split AUROC is the main-table endpoint, and both unseen-entity endpoints appear in the complete results. All six arms and three seeds complete the full protocol.

DrugBAN supplies an additional architecture reference under the uniform multi-laboratory trainer (Bai et al., 2023; DrugBAN contributors, 2026).

4.3 PROTEOMICS-BASED EFFICACY PREDICTION

The public ProteinTalks PTPC supplement provides measured proteomic responses and binary efficacy labels (Sun et al., 2026). The task predicts efficacy from a treated condition’s observed 6h/24h proteome and drug representation. We partition the 491-condition source pool, spanning 333 compounds, into 386 fitting and 105 development conditions by compound group. Ten clients each own 49–50 total conditions, with compound-disjoint training and development groups and both classes in every local development set. Local fitting sets contain 35–42 conditions.

The task reference is a ProteinTalks-derived observed-response efficacy head: protein and drug convolutional branches followed by a 64–32–1 classifier, trained from random initialization on observed 6h/24h responses and Morgan features. A 24h linear/Morgan model provides a complementary non-neural reference.

Average precision (AP) is primary; AUROC, binary cross-entropy and accuracy at 0.5 are secondary. The 148-condition test role is evaluated after the development-selected checkpoint is frozen.

4.4 CELL PERTURBATION IDENTIFICATION

Given an observed expression response and five candidate interventions, the model selects the true perturbation by cosine agreement with predicted responses. The three strata cover VCC single-gene perturbations, Norman double-gene overexpression, and Tahoe drugs at a fixed cell context and dose. Whole-intervention train/development/test counts are respectively 60/20/20, 70/20/27 and 180/60/67. Each laboratory owns six/seven/eighteen training interventions and two/two/six development interventions. Assignment balances intervention identities, while retaining the naturally varying number of VCC cells per intervention.

The public CELLxGENE-pretrained scDEBART backbone is frozen and its response MLP is trainable (Sung & Kim, 2026). Three upstream attention masks are implemented as additive blocking masks and verified by response-free synthetic tests. This correction leaves pretrained weights, targets, splits and candidate rosters unchanged; features are then re-extracted for every arm.

Genetic tasks use the released perturbation encoding and exclude directly perturbed genes from loss and ranking coordinates. Tahoe adds fixed Morgan-projection conditioning for chemical identities. The uniform cell objective combines response MSE with 0.1 times a contrastive loss formed from local training conditions.

Primary scoring is intervention-macro Top-1, with a 20% random-choice expectation; micro Top-1, MRR, Top-3 and cosine-logit cross-entropy are secondary. All arms receive the same five candidate interventions for each response.

5. RESULTS

5.1 MAIN HELD-OUT PERFORMANCE

Table 1 presents the three biological task families as five explicit benchmark columns: one DTI endpoint, one proteomic efficacy endpoint and three cell-perturbation strata. It compares a fixed recipe and feedback-free direct search using one laboratory with the complete AI4AI4Cell system using ten laboratories. Task-reference models are TAPB, the ProteinTalks-derived efficacy head and mask-corrected scDEBART adapters. Every cell reports post-selection held-out mean $\pm$ sample standard deviation over the three completed seeds 42–44. This standard deviation measures variation across training/search runs on fixed data partitions, not a confidence interval or variation across independent datasets. The final column is computed within each seed across the five displayed endpoints before its mean and standard deviation are taken; Appendix A provides the complete six-arm factorial results and secondary metrics.

Method	DTI	Proteomics	Cell perturbation			Overall
	TAPB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$	Mean 5 endpoints
Fixed recipe (1 lab)	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00	36.55 $\pm$ 0.82
Direct optimization (1 lab)	<u>89.89</u> $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21	32.14 $\pm$ 2.02
AI4AI4Cell (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59	41.39 $\pm$ 0.60

Table 1: Main held-out comparison across three biological task families and five endpoints. Fixed recipe and feedback-free direct search use one laboratory; AI4AI4Cell uses ten. PTPC denotes observed-response proteomic efficacy. Values are mean $\pm$ sample SD across repeated training/search seeds, multiplied by 100; SD is not a confidence interval. Overall is computed per common seed before taking its mean and SD. Exact seed counts: DTI 3, Proteomics 3, VCC 3, Norman 3, Tahoe 3; Overall 3. Bold and underline rank the displayed means. All six factorial arms and secondary metrics appear in Appendix A.

Benchmark	Role	Direct	Loop	Gain	W/T/L
DTI: TAPB	Dev. replay	93.63	94.02	+0.39	2/1/0
Proteomics: efficacy	Held-out	37.47	37.47	+0.00	0/3/0
Cell: VCC single-gene	Held-out	20.01	20.01	+0.00	0/3/0
Cell: Norman double-gene	Held-out	11.48	22.32	+10.84	3/0/0
Cell: Tahoe drug	Held-out	20.90	22.89	+1.99	1/2/0

Table 2: Compute-matched feedback isolation, averaged over three seeds and reported in percentage points. Held-out arms each use 800 full-client rounds and 160 development measurements. Gain is loop minus direct; W/T/L counts seed-level wins, ties and losses. The held-out total is 4/8/0, with Norman 95% CI [4.54, 17.73] and Tahoe CI [−1.00, 7.46]. DTI reports a conservative development-trajectory replay; no DTI held-out claim is made.

AI4AI4Cell is the highest-scoring displayed configuration on four of five endpoints and raises the descriptive five-endpoint mean to 41.39, versus 36.55 for the one-laboratory fixed recipe and 32.14 for one-laboratory direct optimization. TAPB reaches 93.76% random-test AUROC versus 89.89% for direct optimization, while proteomic AP reaches 36.99% versus 20.13%. Across VCC, Norman and Tahoe, AI4AI4Cell changes macro Top-1 by +10.45, +8.10 and +6.97 points relative to direct optimization. The complete system therefore combines strong results in molecular interaction, protein-response and cellular-perturbation settings within one research interface. The remaining exception is Norman, where the one-laboratory fixed recipe scores 25.56% and AI4AI4Cell scores 22.25%.

5.2 SEPARATING PARTICIPATION, SEARCH AND FEEDBACK

Table 2 isolates the value of iterative evidence under matched data, candidate and compute budgets. It compares uniform feedback-free allocation with the screening–promotion–validation loop; both arms use ten laboratories and the same candidate slate.

The compute-matched loop is never worse on the 12 newly executed held-out task–seed evaluations: it wins four and ties eight. The largest effect is on Norman double-gene perturbations, where the held-out macro Top-1 mean increases from 11.48% to 22.32%, a +10.84-point effect with hierarchical paired-bootstrap 95% CI [4.54, 17.73]. Tahoe improves by +1.99 points on average, with one win and two ties; VCC and proteomics tie exactly. On DTI, a conservative replay of previously completed development trajectories gives the feedback-free arm slightly more candidate-rounds (525 versus 520), yet the loop wins two seeds, ties one and adds 0.39 AUROC points on average. This replay is reported as development evidence rather than a new held-out result.

The gain comes from allocation rather than additional designs or measurements. Uniform search spends the same 80-round budget on every candidate. The loop uses short trajectories to retain both strong current candidates and candidates that are still improving, then concentrates full validation on that promoted set. On Norman, the promoted low-learning-rate residual design continues improving after the uniform arm stops; the resulting checkpoint improves held-out performance in all three

seeds. The exact ties on VCC and proteomics show that racing preserves the strongest uniform-search solution when additional validation is unnecessary.

The original factorial experiment remains useful for separating participation and design search. Holding the research loop fixed, ten-laboratory participation adds +3.87 AUROC points for DTI, +16.85 AP points for proteomics, and +11.32, +8.10 and +6.97 Top-1 points for VCC, Norman and Tahoe. Appendix A reports the complete six-arm results, proposal prefixes, secondary metrics, DTI unseen-entity endpoints and paired bootstrap intervals.

5.3 EXECUTION AND RESEARCH-HISTORY EVIDENCE

All branches share the proposer, design library, fitting schedule, aggregator and finalizer. Their paired first proposal is identical; subsequent loop proposals additionally receive the accumulated evidence cards. Checkpoint seals, saved predictions and independent score recomputation connect every reported value to an executed run. The histories retain both successful and unsuccessful directions, so a selected design can be traced back through its hypothesis, configuration, development trajectory and retention decision.

Implementation checks exercise the same interfaces used by the scientific runs. The common trainer applies server momentum to parameters while aggregating model buffers by type. Five response-free tests verify the cell attention masks, including causal direction, padding exclusion, unchanged pretrained weights and BF16 execution, after which every cell arm is evaluated on the corrected features.

6. DISCUSSION

AI4AI4Cell turns distributed biological measurements into two coupled research signals. Local training produces parameter updates for a shared predictor; local development produces evidence cards that shape the next executable proposal. The same task contract, proposal schema, experiment memory and finalizer operate across DTI, proteomics and cell perturbation, making the outer research process portable even though the biological inputs, adapters and metrics differ.

The experiments identify two complementary sources of improvement. Broader laboratory participation consistently strengthens the complete system across all five endpoints, demonstrating the value of combining distributed fitting and selection evidence. The compute-matched racing study then isolates persistent feedback from that participation effect: across 12 newly executed held-out trajectories it produces four wins, eight ties and no losses, including a +10.84-point mean Norman improvement with a positive hierarchical bootstrap interval. The DTI development replay supplies the same directional result under a slightly conservative compute comparison. Together, these results show that evidence cards are useful not only as an experiment record but also as a control signal for deciding which computational designs merit full validation.

The current benchmark establishes a reproducible starting point for that expansion. The next experimental stage can add independently collected laboratory cohorts, client rotations and same-data centralized references; broaden the editable interface from optimizer and residual-head configurations to model modules and data transformations; and combine the worker interface with secure aggregation or differential privacy. These additions preserve the central abstraction introduced here: laboratories contribute to both parameter learning and research-direction selection through a common executable evidence loop.

7. CONCLUSION

AI4AI4Cell couples client-local parameter fitting with an evidence-guided research harness, enabling biological measurements held by multiple laboratories to improve both a shared predictor and its executable design. Across DTI, observed-response proteomics and three cell-perturbation strata, the complete system leads four of five main-table endpoints. The factorial evaluation attributes the broadest gains to multi-laboratory participation, while the compute-matched racing experiment shows that persistent feedback can also improve how a fixed training budget is allocated: four held-out wins, eight ties and no losses, with a +10.84-point mean gain on Norman. A shared proposal schema,

evidence-card history and laboratory interface make this process reproducible across task families and provide a practical foundation for collaborative, self-improving biological AI.

REPRODUCIBILITY AND RESPONSIBLE USE

The uniform implementation is versioned as `extensions/unified_bio_20260918`. Definition files bind the common core, task adapters, proposer, runner, data manifest, training seed and budget. For every proposal, the artifact records the structured request, executable configuration, accepted/rejected/failed outcome, training progress, selected checkpoint and aggregate development report. The racing study additionally records the complete short-run slate, promotion decision, full-validation set, equal-compute assertion and independently rescored held-out predictions. The manuscript snapshot records source hashes and the executed scalar results used to generate its tables.

The three task adapters use released biological resources under their source access conditions. Raw response arrays and checkpoints remain in the experimental workspace; the writing repository contains aggregate results, provenance records and table-generation code. The ten workers simulate laboratory data holders on one host and communicate through the same update/evidence interfaces that can be deployed across institutions. The present implementation uses standard sample-weighted aggregation; secure aggregation and differential privacy can be attached to this interface in deployments requiring formal protection.

ETHICS STATEMENT

This retrospective computational study collects no new biological measurements and enrolls no human participants. The ten-client partitions simulate institutional data boundaries. Predictions are research outputs for benchmark evaluation and require independent experimental validation before biological or therapeutic use.

AI USE STATEMENT

The uniform six-arm study uses local Qwen2.5-7B-Instruct for both direct and loop proposal generation. AI coding assistants supported implementation, data-quality checks, analysis, visualization, literature retrieval and manuscript editing. Reported measurements come from executed scorers and traceable artifacts. Human authors reviewed the AI-assisted work and take responsibility for the paper.

REFERENCES

- Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 2025. doi: 10.1038/s41592-025-02772-6. URL <https://doi.org/10.1038/s41592-025-02772-6>.
- Peizhen Bai, Filip Miljković, Bino John, and Haiping Lu. Interpretable bilinear attention network with domain adaptation improves drug-target prediction. *Nature Machine Intelligence*, 5:126–136, 2023. doi: 10.1038/s42256-022-00605-1. URL <https://doi.org/10.1038/s42256-022-00605-1>.
- Mohammad Bakhtiari, Stefan Bonn, Fabian Theis, Olga Zolotareva, and Jan Baumbach. FedscGen: privacy-preserving federated batch effect correction of single-cell RNA sequencing data. *Genome Biology*, 26:216, 2025. doi: 10.1186/s13059-025-03684-6. URL <https://doi.org/10.1186/s13059-025-03684-6>.
- Jiayuan Ding, Jianhui Lin, Ziyang Miao, Nils Mechtel, Shiyu Jiang, Yixin Wang, Zhaoyu Fang, Jorge D. Martin-Rufino, Chen Weng, Reuben Saunders, Weize Xu, Jonathan S. Weissman, Min Li, Jiliang Tang, Wei Ouyang, Yuancheng Ryan Lu, and Xiaojie Qiu. Predictive single cell foundation model for gene regulation and aging with privacy-preserving tabular learning, 2026. URL <https://arxiv.org/abs/2607.19400v1>. Version 1.
- DrugBAN contributors. DrugBAN: Official training implementation, 2026. URL <https://github.com/peizhenbai/DrugBAN/tree/9923f8c99959e00263103ff9ac61ba0eaccc8e02>. Commit 9923f8c99959e00263103ff9ac61ba0eaccc8e02.

- GAIR-NLP DrugEvolve contributors. DrugEvolve: Official source and applications, 2026. URL <https://github.com/GAIR-NLP/DrugEvolve/tree/e945c7d0da28ddce73904419116a51e03921e6e6>. Commit e945c7d0da28ddce73904419116a51e03921e6e6.
- Zhengyao Jiang, Dominik Schmidt, Dhruv Srikanth, Dixing Xu, Ian Kaplan, Deniss Jacenko, and Yuxiang Wu. AIDE: AI-driven exploration in the space of code, 2025. URL <https://arxiv.org/abs/2502.13138v1>. Version 1, 18 February 2025.
- Mikhail Khodak, Renbo Tu, Tian Li, Liam Li, Maria-Florina Balcan, Virginia Smith, and Ameet Talwalkar. Federated hyperparameter tuning: Challenges, baselines, and connections to weight-sharing. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://papers.neurips.cc/paper_files/paper/2021/hash/a0205b87490c847182672e8d371e9948-Abstract.html.
- Haoyuan Li, Mathias Funk, and Aaqib Saeed. Helmsman: Autonomous synthesis of federated learning systems via collaborative LLM agents, 2025. URL <https://arxiv.org/abs/2510.14512v2>. Version 2, 19 December 2025.
- Gaoming Lin, Xin Zhang, Zhonghao Ren, Quan Zou, Prayag Tiwari, Changjun Zhou, and Yijie Ding. TAPB: an interventional debiasing framework for alleviating target prior bias in drug-target interaction prediction. *Nature Communications*, 16:10867, 2025. doi: 10.1038/s41467-025-66915-1. URL <https://doi.org/10.1038/s41467-025-66915-1>.
- Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery, 2024. URL <https://arxiv.org/abs/2408.06292v1>. Version 1, 12 August 2024.
- Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pp. 1273–1282, 2017. URL <https://proceedings.mlr.press/v54/mcmahan17a.html>.
- Nils Mechtel, Wei Ouyang, and Chiron Platform contributors. Chiron Platform: Federated single-cell training and agent interfaces, 2026. URL <https://github.com/aristoteleo/chiron/tree/3241b35c8b516c126aa68da450708de3da717ee0>. Commit 3241b35c8b516c126aa68da450708de3da717ee0.
- Micah J. Sheller, Brandon Edwards, G. Anthony Reina, Jason Martin, Sarthak Pati, Aikaterini Kotrotsou, Mikhail Milchenko, Weilin Xu, Daniel Marcus, Rivka R. Colen, and Spyridon Bakas. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports*, 10:12598, 2020. doi: 10.1038/s41598-020-69250-1. URL <https://doi.org/10.1038/s41598-020-69250-1>.
- Rui Sun, Liujia Qian, Yongge Li, Tong Liu, Honghan Cheng, Xuedong Zhang, Xueya Zhou, Yuecheng Zhan, Guangmei Zhang, Zhengchao Luo, Kunpeng Ma, Chunlong Wu, Dongchen Ji, Zhangzhi Xue, Hongxue Meng, Yuhang Xiang, Dingwei Lei, Qianhe Zhou, Wenbin Hu, Yuhang Deng, Lingling Tan, Qi Xiao, Zhiwei Liu, Lei Zeng, Liqin Qian, Xuan Zheng, Qiong Hu, Nianzi Luo, Weinan E, Peijie Zhou, Han Wen, Yi Zhu, and Tiannan Guo. An operational perturbation proteomics-based virtual cell model. *Nature*, 2026. doi: 10.1038/s41586-026-11001-9. URL <https://doi.org/10.1038/s41586-026-11001-9>. Published online 9 September 2026.
- Jieun Sung and Wankyu Kim. scDEBART: Predicting in silico single-cell perturbation responses via large-scale differential expression learning. In *International Conference on Machine Learning*, 2026. URL <https://openreview.net/forum?id=pJyidZg93y>.
- Shuang Wang, Bochen Shen, Lanting Guo, Mengqi Shang, Jinze Liu, Qi Sun, and Bairong Shen. scFed: federated learning for cell type classification with scRNA-seq. *Briefings in Bioinformatics*, 25(1):bbad507, 2024. doi: 10.1093/bib/bbad507. URL <https://doi.org/10.1093/bib/bbad507>.

Zhimeng Zhou, Yang Nan, Minjie Mou, Yuntao Qian, Yixiu Liu, Zhengyu Zuo, Hao Yang, Weixian Xu, Bo Li, Wanghao Jiang, Yanlin Ren, Yang Liao, Yimeng Wang, Yinghong Li, Qingxia Yang, Zhiheng Xi, Tiantian Mi, Huaicheng Sun, Pengfei Liu, and Feng Zhu. An AI system for autonomous algorithm evolution in drug development. *bioRxiv*, 2026. doi: 10.64898/2026.08.16.745117. URL <https://doi.org/10.64898/2026.08.16.745117>. Version 1, posted 20 August 2026.

APPENDIX A. UNIFORM COLLABORATIVE RESEARCH PROTOCOL

Versioned implementation. The reported study uses `core_v3.py`, `run_v3.py` and `research_v2.py`. All task families execute the same optimization, aggregation, proposal and retention source through task-specific adapters. TAPB workers use the TAPB environment; Qwen, proteomics and cell adapters use the LDA environment. Each scientific fit starts from its task-specific public or random initialization, with no reuse of earlier task-trained weights.

Complete design library. Each identifier below compiles to one executable candidate. D00 is the fixed starting recipe; D01–D11 are available to both direct and loop search without within-arm repetition. Six proposal slots define the common budget.

Design	Learning rate	Weight decay	Momentum	Residual adapter
D00	0.001	0.001	0	no
D01	0.0001	0.001	0	no
D02	0.01	0.001	0	no
D03	0.001	0	0	no
D04	0.001	0.01	0	no
D05	0.001	0.001	0.9	no
D06	0.001	0.001	0	yes
D07	0.001	0.001	0.9	yes
D08	0.0001	0.001	0	yes
D09	0.0001	0.001	0.9	yes
D10	0.01	0.01	0	no
D11	0.001	0.01	0	yes

Proposal generation and evidence memory. The proposer uses bfloat16 inference, temperature 0.7, top- p 0.9 and at most 512 new tokens. Each slot uses a deterministic seed derived from the training seed. Direct and loop arms share the same first proposal. Later direct proposals receive the incumbent and tried design identifiers. Later loop proposals additionally receive an ordered list of evidence cards, each containing the candidate configuration, aggregate best development metric and loss, best round, first/last training and validation records, and final status. A bounded second generation attempt repairs schema or novelty violations.

Every proposal has the following executable JSON shape:

```
{
  "hypothesis": "one testable claim about an editable factor",
  "experiment": "the controlled change and incumbent comparison",
  "expected_effect": "the expected metric and diagnostic response",
  "design_id": "D00--D11"
}
```

The compiler validates `design_id` against the table above. The finalizer then attaches executed measurements and a retention decision to the proposal, creating the evidence card used in the next iteration.

Executed loop trajectories. Three ten-laboratory traces illustrate how the common finalizer operates. Values are aggregate development measurements on the 0–1 primary-metric scale, and proposal slots are one-indexed.

Study	Selected design	Primary	Loss	First direct / loop slot
PTPC, seed 42	D04	0.665833	0.521161	5 / 5
VCC, seed 43	D00	0.240350	1.607792	initial baseline
Norman, seed 42	D08	0.519000	1.538505	3 / 2

PTPC selects stronger weight decay through the BCE tie-break at equal AP. VCC retains the initial design after six alternatives. On Norman, the loop reaches the winning lower-learning-rate residual

Labs	Method	DTI Proteomics		Cell perturbation		
		AUROC ↑	AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
1	Fixed model	82.83±1.22	25.05±4.56	29.94±0.00	25.56 ±0.00	19.40±0.00
1	Direct	89.89±1.02	20.13±0.07	21.09±4.38	14.15±0.00	15.42±6.21
1	Our loop	89.89±1.02	20.13±0.07	20.22±2.22	14.15±0.00	15.42±6.21
10	Fixed model	91.80±2.55	37.07 ±0.08	31.54 ±0.08	15.48±0.00	25.87 ±1.72
10	Direct	93.76 ±1.11	36.99±0.22	31.54 ±0.08	22.25±0.45	22.39±2.59
10	Our loop	93.76 ±1.11	36.99±0.22	31.54 ±0.08	22.25±0.45	22.39±2.59
Completed seeds ($n/3$)		3/3	3/3	3/3	3/3	3/3

Table A.1: Six-arm ablation of the task-reference models. Entries are held-out mean $\pm$ sample SD across completed training/search seeds, multiplied by 100. The data partitions remain fixed; this SD measures run-to-run variation, not dataset uncertainty. DTI uses the random held-out endpoint. Bold marks all displayed maxima.

design one slot before direct search. Across all 15 ten-laboratory task–seed trajectories, the two-proposal loop is better in two, tied in thirteen and worse in none; the mean development-primary difference is +0.78 percentage points. At six proposals, direct and loop select the same checkpoints, reflecting coverage saturation in the twelve-design library.

Compute-matched evidence racing. The follow-up feedback-isolation protocol uses a ten-design fractional-factorial slate and newly executed seeds 53–55. Direct search assigns 80 rounds to every design. The loop assigns 20 screening rounds to every design, stratifies candidates by learning rate, and promotes both the strongest current candidate and the fastest-improving candidate within each stratum. Six promoted candidates restart from the common initialization for 100-round validation. Consequently, each arm executes 800 full-client rounds and 160 development measurements over the same ten hypotheses. The promotion record is finalized before any held-out response is read.

Across the resulting 12 proteomics/cell held-out comparisons, racing gives four wins, eight ties and no losses. Norman improves by 10.84 Top-1 points on average, with seed-level effects of +10.00, +11.70 and +10.81 and a hierarchical paired-bootstrap 95% interval of [4.54, 17.73]. Tahoe gives one +5.97-point seed and two ties; VCC and PTPC tie in all three seeds. Independent rescoring of every saved prediction reproduces the reported metrics exactly. A TAPB companion replay applies the same screen/promote principle to the seven-design intersection of existing traces: direct receives 525 candidate-rounds, the loop receives 520, and the development AUROC effect is +0.39 points on average (two wins, one tie).

Client and split integrity. DTI clients contain 4,932–5,058 training and 517–643 development pairs. Compound identities do not cross clients. The fixed test roles contain 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein pairs. PTPC partitions 333 compound groups into 386 fitting and 105 development conditions, with 35–42 fitting conditions per client; its frozen test role contains 148 conditions. VCC, Norman and Tahoe use whole-intervention splits of 60/20/20, 70/20/27 and 180/60/67. Their ten clients own equal numbers of training and development intervention identities.

Proteomic preprocessing. For each 6h and 24h profile, the PTPC adapter applies log1p, within-profile min–max normalization over finite observations, zero filling and unit L2 normalization. The 2,048-bit Morgan vector is separately L2-normalized and scaled by 0.5. Concatenating the two 5,171-dimensional proteomic blocks with the drug vector produces a 12,390-dimensional float32 input. The ProteinTalks-derived efficacy head contains 464,193 parameters at D00. Every arm uses identical inputs, initialization policy, 100-round budget and checkpoint rule.

Full factorial results. The following tables report all six held-out arms, complete method contrasts, proposal-prefix analysis, secondary metrics, uncertainty and all three DTI test roles. Bold marks the best displayed value within each task and metric, including ties. Higher is better except BCE and cross-entropy.

Held-out contrast	DTI	Proteomics	VCC	Norman	Tahoe
Loop – fixed (1 lab)	+7.06	-4.91	-9.71	-11.41	-3.98
Loop – fixed (10 labs)	+1.96	-0.08	+0.00	+6.77	-3.48
Loop – direct (1 lab)	+0.00	+0.00	-0.87	+0.00	+0.00
Loop – direct (10 labs)	+0.00	+0.00	+0.00	+0.00	+0.00
10 – 1 labs (both fixed)	+8.97	+12.02	+1.61	-10.07	+6.47
10 – 1 labs (both direct)	+3.87	+16.85	+10.45	+8.10	+6.97
10 – 1 labs (both loop)	+3.87	+16.85	+11.32	+8.10	+6.97
Completed seeds ($n/3$)	3/3	3/3	3/3	3/3	3/3

Table A.2: Complete decomposition into design-search improvement (loop minus fixed), feedback-specific improvement (loop minus matched-budget direct), and laboratory-participation improvement (ten minus one lab with loop fixed). Mean paired held-out differences are in percentage points over the three seeds and are computed before rounding.

Proposal budget	Loop better	Tie	Direct better	Mean Δ primary
	task–seed	trajectories		points
1	0	15	0	+0.00
2	2	13	0	+0.78
3	0	14	1	-0.04
4	0	15	0	+0.00
5	0	14	1	-0.02
6	0	15	0	+0.00

Table A.3: Executed-prefix analysis of all 15 ten-laboratory development trajectories at common proposal budgets. At two proposals, feedback-guided search improves TAPB seed 44 by 4.84 points and Norman seed 42 by 6.90 points, with thirteen ties and no regressions.

Table A.4: Held-out secondary metrics for the task-reference models, averaged over seeds 42–44. Bold indicates all tied maxima, or minima for loss.

Method	Labs	Metric 1	Metric 2	Metric 3	Metric 4
DTI ($n = 3$)		AUROC	AP	MCC	BCE
Fixed model	1	0.8283	0.8157	0.4995	0.5237
Direct	1	0.8989	0.8777	0.6571	0.6102
Our loop	1	0.8989	0.8777	0.6571	0.6102
Fixed model	10	0.9180	0.8936	0.7239	0.3654
Direct	10	0.9376	0.9273	0.7371	0.3659
Our loop	10	0.9376	0.9273	0.7371	0.3659
Proteomics ($n = 3$)		AP	AUROC	BCE	Acc.
Fixed model	1	0.2505	0.6549	0.4265	0.8514
Direct	1	0.2013	0.6180	0.4434	0.8176
Our loop	1	0.2013	0.6180	0.4434	0.8176
Fixed model	10	0.3707	0.7406	0.3978	0.8514
Direct	10	0.3699	0.7387	0.3983	0.8514
Our loop	10	0.3699	0.7387	0.3983	0.8514
VCC ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.3361	0.5653	0.7389	1.5451
Direct	1	0.2148	0.4710	0.6278	1.6082
Our loop	1	0.2074	0.4694	0.6426	1.6086
Fixed model	10	0.3398	0.5634	0.6944	1.5814
Direct	10	0.3398	0.5634	0.6944	1.5814
Our loop	10	0.3398	0.5634	0.6944	1.5814
Norman ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.2556	0.4850	0.6393	1.5874
Direct	1	0.1415	0.4212	0.5835	1.6039
Our loop	1	0.1415	0.4212	0.5835	1.6039
Fixed model	10	0.1548	0.4526	0.6563	1.5806
Direct	10	0.2225	0.5007	0.7096	1.5883
Our loop	10	0.2225	0.5007	0.7096	1.5883
Tahoe ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.1940	0.4500	0.5821	1.6308
Direct	1	0.1542	0.4260	0.5672	1.6190
Our loop	1	0.1542	0.4260	0.5672	1.6190
Fixed model	10	0.2587	0.5020	0.6517	1.6061
Direct	10	0.2239	0.4799	0.6169	1.6075
Our loop	10	0.2239	0.4795	0.6119	1.6075

Table A.5: Paired cluster-bootstrap 95% intervals for each seed and main contrast, in percentage points. Intervals condition on the selected trained models; DTI random-endpoint rows resample drug clusters.

Task	Contrast	Seed	Difference	95% interval
DTI	Loop – fixed (10 labs)	42	+0.33	[-0.06, +0.69]
DTI	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	42	+12.51	[+11.35, +13.71]
DTI	10 – 1 labs (both direct)	42	+5.78	[+5.02, +6.57]
DTI	10 – 1 labs (both loop)	42	+5.78	[+5.02, +6.57]
DTI	Loop – fixed (10 labs)	43	+0.54	[-0.08, +1.11]
DTI	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]

Task	Contrast	Seed	Difference	95% interval
DTI	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	43	+7.80	[+6.82, +8.82]
DTI	10 – 1 labs (both direct)	43	+1.65	[+0.90, +2.38]
DTI	10 – 1 labs (both loop)	43	+1.65	[+0.90, +2.38]
DTI	Loop – fixed (10 labs)	44	+5.01	[+4.25, +5.71]
DTI	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	44	+6.60	[+5.61, +7.74]
DTI	10 – 1 labs (both direct)	44	+4.17	[+3.45, +4.93]
DTI	10 – 1 labs (both loop)	44	+4.17	[+3.45, +4.93]
Proteomics	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	42	+13.16	[-0.39, +26.02]
Proteomics	10 – 1 labs (both direct)	42	+16.91	[+3.37, +31.02]
Proteomics	10 – 1 labs (both loop)	42	+16.91	[+3.37, +31.02]
Proteomics	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	43	+15.98	[+1.29, +29.00]
Proteomics	10 – 1 labs (both direct)	43	+17.03	[+3.34, +31.44]
Proteomics	10 – 1 labs (both loop)	43	+17.03	[+3.34, +31.44]
Proteomics	Loop – fixed (10 labs)	44	-0.24	[-3.40, +2.33]
Proteomics	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	44	+6.93	[-3.43, +15.60]
Proteomics	10 – 1 labs (both direct)	44	+16.63	[+3.02, +31.08]
Proteomics	10 – 1 labs (both loop)	44	+16.63	[+3.02, +31.08]
VCC	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	42	+1.70	[-5.36, +8.58]
VCC	10 – 1 labs (both direct)	42	+9.90	[-2.88, +23.04]
VCC	10 – 1 labs (both loop)	42	+9.90	[-2.88, +23.04]
VCC	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	43	+4.83	[-5.50, +16.49]
VCC	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	43	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	43	+15.08	[+2.94, +27.78]
VCC	10 – 1 labs (both loop)	43	+10.24	[-2.74, +23.49]
VCC	Loop – fixed (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	44	-7.44	[-17.23, +1.33]
VCC	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	44	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	44	+6.38	[-4.38, +19.13]
VCC	10 – 1 labs (both loop)	44	+13.82	[+2.54, +26.34]
Norman	Loop – fixed (10 labs)	42	+7.26	[-9.41, +25.41]
Norman	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	42	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	42	+8.59	[-8.30, +26.00]
Norman	10 – 1 labs (both loop)	42	+8.59	[-8.30, +26.00]
Norman	Loop – fixed (10 labs)	43	+6.37	[-11.12, +24.81]

Task	Contrast	Seed	Difference	95% interval
Norman	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	43	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	43	+7.70	[-9.33, +25.85]
Norman	10 – 1 labs (both loop)	43	+7.70	[-9.33, +25.85]
Norman	Loop – fixed (10 labs)	44	+6.67	[-10.37, +24.81]
Norman	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	44	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	44	+8.00	[-8.52, +25.93]
Norman	10 – 1 labs (both loop)	44	+8.00	[-8.52, +25.93]
Tahoe	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	42	+4.48	[-5.97, +14.93]
Tahoe	10 – 1 labs (both direct)	42	+10.45	[-1.49, +22.39]
Tahoe	10 – 1 labs (both loop)	42	+10.45	[-1.49, +22.39]
Tahoe	Loop – fixed (10 labs)	43	-2.99	[-7.46, +0.00]
Tahoe	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	43	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	43	+13.43	[+1.49, +23.88]
Tahoe	10 – 1 labs (both loop)	43	+13.43	[+1.49, +23.88]
Tahoe	Loop – fixed (10 labs)	44	-7.46	[-19.40, +4.48]
Tahoe	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	44	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	44	-2.99	[-17.91, +11.94]
Tahoe	10 – 1 labs (both loop)	44	-2.99	[-17.91, +11.94]

Table A.6: All three frozen TAPB held-out endpoints under the committed common protocol. Scores are completed-seed means multiplied by 100; N/A means no complete independently verified six-arm evaluation. The main DTI endpoint is random held-out.

Endpoint	Method	Labs	AUROC	AP
Random	Fixed model	1	82.83	81.57
Random	Direct	1	89.89	87.77
Random	Our loop	1	89.89	87.77
Random	Fixed model	10	91.80	89.36
Random	Direct	10	93.76	92.73
Random	Our loop	10	93.76	92.73
Unseen drug	Fixed model	1	81.96	85.78
Unseen drug	Direct	1	88.39	89.35
Unseen drug	Our loop	1	88.39	89.35
Unseen drug	Fixed model	10	90.06	90.40
Unseen drug	Direct	10	91.65	92.58
Unseen drug	Our loop	10	91.65	92.58
Unseen protein	Fixed model	1	76.40	78.75
Unseen protein	Direct	1	83.26	80.53
Unseen protein	Our loop	1	83.26	80.53
Unseen protein	Fixed model	10	85.80	78.92

Endpoint	Method	Labs	AUROC	AP
Unseen protein	Direct	10	87.63	85.07
Unseen protein	Our loop	10	87.63	85.07

Table A.7: Post-hoc TAPB paired AUROC/AP differences and 95% intervals from 1,000 fixed-seed cluster-bootstrap draws. Random and unseen-drug endpoints use drug clusters; unseen-protein uses protein clusters. All three endpoints, six arms and prespecified contrasts are shown.

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Random / 42	Loop — fixed (10 labs)	+0.33	[-0.06, +0.69]	+1.40	[+0.76, +2.05]
Random / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	10 — 1 labs (fixed)	+12.51	[+11.35, +13.71]	+10.56	[+9.10, +12.11]
Random / 42	10 — 1 labs (direct)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	10 — 1 labs (loop)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	Our harness — fixed (1 lab)	+12.84	[+11.82, +14.01]	+11.96	[+10.56, +13.45]
Random / 42	Our harness — direct (1 lab)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Unseen drug / 42	Loop — fixed (10 labs)	+0.10	[-0.47, +0.67]	+1.06	[+0.42, +1.66]
Unseen drug / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	10 — 1 labs (fixed)	+12.30	[+10.77, +13.98]	+7.17	[+5.82, +8.72]
Unseen drug / 42	10 — 1 labs (direct)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	10 — 1 labs (loop)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	Our harness — fixed (1 lab)	+12.40	[+10.94, +13.85]	+8.22	[+6.97, +9.66]
Unseen drug / 42	Our harness — direct (1 lab)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen protein / 42	Loop — fixed (10 labs)	+0.03	[-5.53, +4.28]	+4.12	[-4.10, +8.22]
Unseen protein / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	10 — 1 labs (fixed)	+20.02	[+0.16, +63.83]	+8.14	[-8.27, +51.77]
Unseen protein / 42	10 — 1 labs (direct)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	10 — 1 labs (loop)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	Our harness — fixed (1 lab)	+20.05	[+1.84, +61.27]	+12.25	[-3.01, +53.19]
Unseen protein / 42	Our harness — direct (1 lab)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Random / 43	Loop — fixed (10 labs)	+0.54	[-0.08, +1.11]	+0.44	[-0.54, +1.44]
Random / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	10 — 1 labs (fixed)	+7.80	[+6.82, +8.82]	+7.68	[+6.32, +9.13]
Random / 43	10 — 1 labs (direct)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	10 — 1 labs (loop)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	Our harness — fixed (1 lab)	+8.35	[+7.29, +9.37]	+8.12	[+6.53, +9.76]
Random / 43	Our harness — direct (1 lab)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Unseen drug / 43	Loop — fixed (10 labs)	-0.51	[-1.44, +0.29]	-1.10	[-2.16, -0.18]
Unseen drug / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	10 — 1 labs (fixed)	+6.68	[+5.33, +8.25]	+4.66	[+3.33, +6.36]
Unseen drug / 43	10 — 1 labs (direct)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	10 — 1 labs (loop)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	Our harness — fixed (1 lab)	+6.17	[+4.85, +7.48]	+3.57	[+2.10, +5.09]
Unseen drug / 43	Our harness — direct (1 lab)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen protein / 43	Loop — fixed (10 labs)	-0.96	[-7.24, +3.07]	-1.83	[-21.55, +7.54]
Unseen protein / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	10 — 1 labs (fixed)	+4.79	[-6.61, +28.12]	+0.41	[-16.48, +31.69]
Unseen protein / 43	10 — 1 labs (direct)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	10 — 1 labs (loop)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	Our harness — fixed (1 lab)	+3.83	[-5.83, +22.32]	-1.42	[-14.49, +16.88]
Unseen protein / 43	Our harness — direct (1 lab)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Random / 44	Loop — fixed (10 labs)	+5.01	[+4.25, +5.71]	+8.28	[+7.07, +9.56]
Random / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	10 — 1 labs (fixed)	+6.60	[+5.61, +7.74]	+5.12	[+3.71, +6.80]
Random / 44	10 — 1 labs (direct)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	10 — 1 labs (loop)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	Our harness — fixed (1 lab)	+11.60	[+10.60, +12.65]	+13.40	[+11.88, +15.08]
Random / 44	Our harness — direct (1 lab)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Unseen drug / 44	Loop — fixed (10 labs)	+5.18	[+4.09, +6.19]	+6.57	[+5.28, +7.85]
Unseen drug / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	10 — 1 labs (fixed)	+5.32	[+3.81, +6.97]	+2.04	[+0.38, +3.90]
Unseen drug / 44	10 — 1 labs (direct)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen drug / 44	10 — 1 labs (loop)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Unseen drug / 44	Our harness — fixed (1 lab)	+10.50	[+9.20, +11.87]	+8.61	[+7.10, +10.29]
Unseen drug / 44	Our harness — direct (1 lab)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen protein / 44	Loop — fixed (10 labs)	+6.42	[-2.79, +17.37]	+16.15	[-5.79, +26.26]
Unseen protein / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	10 — 1 labs (fixed)	+3.39	[-16.02, +33.81]	-8.03	[-26.82, +16.99]
Unseen protein / 44	10 — 1 labs (direct)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]
Unseen protein / 44	10 — 1 labs (loop)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]
Unseen protein / 44	Our harness — fixed (1 lab)	+9.81	[-2.07, +34.89]	+8.12	[-12.06, +22.71]
Unseen protein / 44	Our harness — direct (1 lab)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]

Uncertainty analysis. The DTI analysis is computed after training and held-out scoring, using paired cluster bootstrap with 1,000 fixed-seed resamples of retained predictions. It groups by released drug identity on random/unseen-drug endpoints and by protein sequence on unseen-protein. Other task intervals use their natural compound or intervention grouping. Resamples are shared within each method contrast. These intervals characterize sample-level variation within each selected fit; between-training-seed variation is reported through the three-seed means and sample SD.

Verification. CPU unit and integration tests cover the common trainer, task adapters, proposal generation, frozen evaluation and cell attention-mask correction. Publication checks bind table cells to completed arms, model versions, checkpoint seals and independent score recomputation. The snapshot includes receipt hashes, aggregate metrics and the complete proposal histories used in this manuscript.